CHAPTER 1

INTRODUCTION

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INTRODUCTION

Human conduct is incredibly affected by their abstract sentiments and convictions, for example, demeanour, feeling, assessment, or slant. The choices we make can be affected by others' impressions of the world to an extensive degree, on the grounds that conveying the others' assessments is wired into every individual normally and portrays us as 'social creatures'. Nowadays, social communication channels like Twitter, Facebook, and YouTube have obtained so much popularity. Opinion mining is the other name of Sentiment Analysis which is under the category of machine learning and data mining. From the use of different social media, opinion mining or sentiment analysis techniques have to start with people ‘s data for the analysis of a different kind of area like politic, economy or biology, etc. Massive amount of information related to distinct individual entities are recorded every day in digital forms. And hence such a fast growth of the field co-exists along with other social media related stuff such a forums discussion, blogs, customer reviews, Twitter and social network sites. Sentiment analysis includes classification of data into various classes like optimistic i.e., good sense or negative i.e., bad sense or neutral i.e. non-effective. Sentiment analysis is the task of perceiving whether a given opinion is positive or negative in general (e.g., a movie review, a person, a political party, or a policy or product feature review. Because of the free format of messages and easy accessibility of microblogging platforms, most of the data on social media are unstructured. When it is necessary to make conclusion or final output, it is important to get Opinions of persons. From different people ‘s experience and reviews which contain important resource. Recent work has been done in promotion and challenging areas with the implementation of opinion mining which serves desires of the individuals. Appropriate training set is required for sentiment analysis for better performance, and accurate dataset for improper analysis of the text. For better means and accuracy, the linguistic analysis is considered Machine learning. With advancements in technology and fields like deep learning, sentiment analysis is becoming more and more common for companies that want to gauge their customers’ sentiments. Sentiment analysis can be defined as analysing the positive or negative sentiment of the customer in text. The contextual analysis of identifying information helps businesses understand their customers’ social sentiment by monitoring online conversations. As customers express their reviews and thoughts about the brand more openly than ever before, sentiment analysis has become a powerful tool to monitor and understand online conversations. Analysing customer feedback and reviews automatically through survey responses or social media discussions allows you to learn what makes your customer happy or disappointed. Further, you can use this analysis to tailor your products and services to meet your customer’s needs and make your brand successful. Recent advancements in machine learning and deep learning have increased the efficiency of sentiment analysis algorithms. You can creatively use advanced artificial intelligence and machine learning tools for doing research and draw out the analysis.

1.1 PROJECT OVERVIEW

The proposal which is used to analyse the tweets. We will be performing sentiment analysis in tweets and determine where it is positive, negative, or neutral. The project can be used by any organization office to review their works or by political leaders or by any others company to review about their products or brands.

The main feature of the project is that it helps to determine the opinion about the peoples on products, government work, politics or any other by analysing the tweets. Our system can train the new tweets taking reference to previously trained tweets. The computed or analysed data will be represented in various diagrams such as Pie chart, Bar graph and word cloud. Sentiment analysis is a specific subtask within the broad area of opinion mining; in short, the classification of texts according to the emotion that the text appears to convey. Sentiment analysis typically classifies texts according to positive, negative, and neutral classifications; so that "*This movie is great!*" is classified as positive, while *"This movie was too long, and I got bored halfway through"* is classified as negative. Opinion mining and sentiment analysis tools, depending on the implementation, often suffer from a few key problems. In particular:

> It is not always clear which of the people or things referenced within a given text are liked or disliked: "*The movie had some decent acting, but I can't forgive the use of Papyrus font for the end credits*".

Sarcasm is difficult to detect: "*Great movie – nearly as much fun as watching paint dry".* Outcomes from a sentiment analysis process may be used to improve recommendation systems; for example, by identifying and reranking contentious products to detect controversy and disagreement. This is potentially a useful ability in automated summarisation tasks, where a range of viewpoints may exist. In the field of social computing, sentiment analysis is envisaged to be useful in supporting collaborative work.

Some of the limitations of sentiment analysis arise from the source material. People are more likely to express positive than negative feelings in social environments and are more likely to express extreme than average feelings.

1.2 EXISTING SYSTEM

A major benefit of social media is that we can see the good and bad things people say about the brand or personality. The bigger your company gets difficult it becomes to keep a handle on how everyone feels about your brand. For large companies with thousands of daily mentions on social media, news sites and blogs, its extremely difficult to do this manually. To combat this problem, sentimental analysis software is necessary. This can be used to evaluate the people's sentiment about brand or personality. Sentimental analysis is used widely in the platforms of business, social media and customer service, marketing sector, reviewer side so on. Multilingual consists of different languages where the classification needs to be done as positive, negative, and neutral. This is highly challenging and comparatively difficult.

1.3 PROPOSED SYSTEM

Twitter plays a vital role in spreading information and influencing people's opinions in a specific direction. As an easy to perform platform, Twitter motivates people to share their thoughts and opinions. This project is made with machine learning techniques to analyse the tweets by Natural Language Processing to process the datasets. And algorithms are used to find the best model by comparing its accuracy.

1.4 PROCESS OF SENTIMENT ANALYSIS

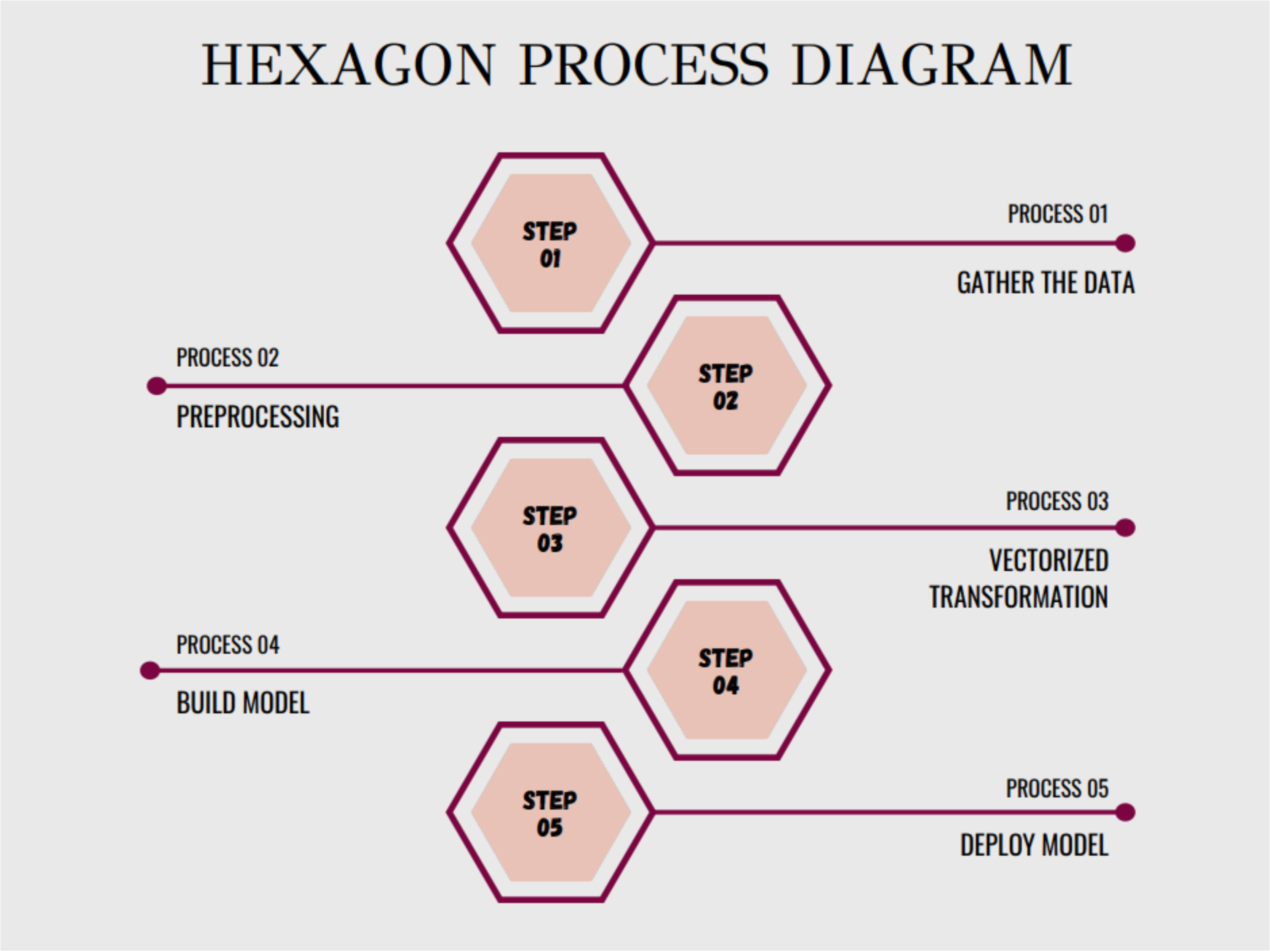

Fig. 1.4 Hexagon process diagram for sentiment analysis

CHAPTER 2

METHODOLOGY

CHAPTER 2
METHODOLOGY

Numerous methodologies are available for opinion mining, but two main groups are used. The problems of SA will be solved by the first group using by implementing the machine learning approach. The second group uses lexicon-based method which is a linguistically-inclined method. In both groups, many techniques exist. From the following way, we can extract the features of text or sentences.

1) N-Gram: Only one word can be taken by one at a time (unigram) or two words (bigram) up to n words as a result. Unigram features cannot be captured by some opinions. For example, this book is fascinating. It is an optimistic comment if in only unigram model it is fascinating to take it together and negative.

2) POS tagging: ―It is the way of words to signify it in content (corpus) as it is linked to its parts of speech in the light of both its definition and its connotation with touching the words. Nouns, pronouns, adjectives, adverbs, etc. are examples of different parts of speech‖.

3) Stemming – In this, eliminating prefixes and suffixes is the main process. For example, ‗running ', ‗sleeping ', ‗ran'can be stemmed from ‗run' and ‗sleep' respectively. It basically helps in Cataloguing but sometimes it also leads to decrease in cataloguing accuracy.

4) Stop words – Stop words are Pronouns (he/she, it), articles (a, an, the), prepositions (above, in, near, under, besides). These words are nothing but offer no or little information about the emotions. On the internet you can access list of stop words. In the pre-processing step, it can be used to remove them.

5) Conjunction handling - In general, there is only one meaning of each sentence at a time. But there are certain available conjunction words like But, And, while, although, however, changes the whole denotation of its sentence. For example, even though the ride was good, but it was not up to my hopes. By using these rules throughput can be amplified by 5% .

6) Negation handling - Negation words like ‗not 'inverts the gist of the whole sentence. For example, the movie was not good as ‗good 'in it which is optimistic but ‗not'upturns the schism to negative.

Data Collection

Consumers usually express their sentiments on public forums like the blogs, discussion boards, product reviews as well as on their private logs – Social network sites like Facebook and Twitter. Opinions and feelings are expressed in different way, with different vocabulary, context of writing, usage of short forms and slang, making the data huge and disorganized. Manual analysis of sentiment data is virtually impossible. Therefore, special programming languages like 'R' are used to process and analyze the data.

Text Preparation

Text preparation is nothing but filtering the extracted data before analysis. It includes identifying and eliminating non-textual content and content that is irrelevant to the area of study from the data.

Sentiment Detection

At this stage, each sentence of the review and opinion is examined for subjectivity. Sentences with subjective expressions are retained and that which conveys objective expressions are discarded. Sentiment analysis is done at different levels using common computational techniques like Unigrams, lemmas, negation and so on.

Sentiment Classification

Sentiments can be broadly classified into two groups, positive and negative. At this stage of sentiment analysis methodology, each subjective sentence detected is classified into groups-positive, negative, good, bad, like, dislike.

Presentation of Output

The main idea of sentiment analysis is to convert unstructured text into meaningful information. After the completion of analysis, the text results are displayed on graphs like pie chart, bar chart and line graphs.

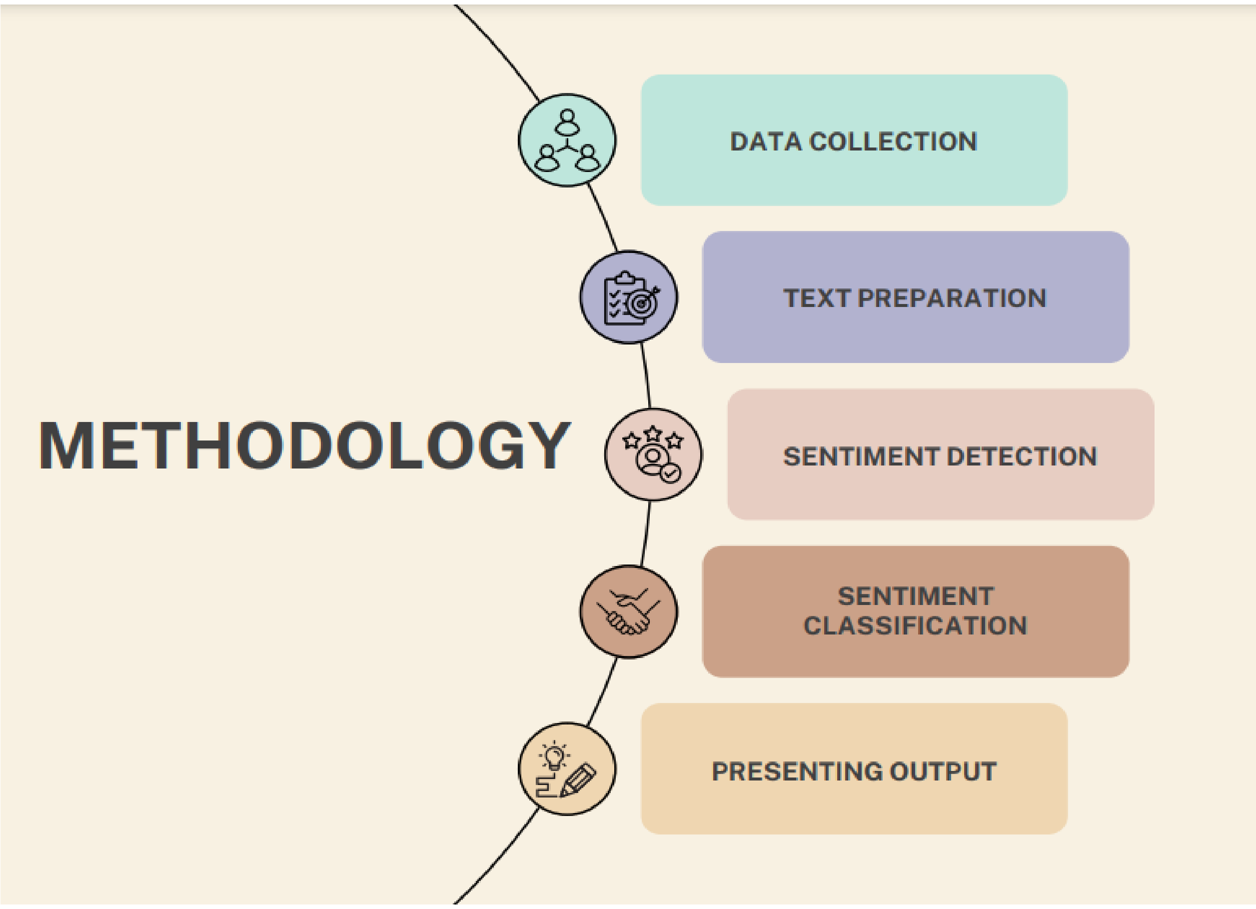

Fig. 2.0 Methodology of project

2.1 PROBLEM DEFINITION

It is often said that if we cannot structure a problem, we probably do not understand the problem. The objective of the definitions is thus to abstract a structure from the complex and intimidating unstructured natural language text. The structure serves as a common framework to unify various existing research directions and enable researchers to design more robust and accurate solution techniques by exploiting the interrelationships of the subproblems. From a practical application point of view, the definitions let practitioners see what subproblems need to be solved in building a sentiment analysis system, how the subproblems are related, and what output should be produced. Unlike factual information, sentiment and opinion have an important characteristic, namely, they are subjective. The subjectivity comes from many sources. First of all, different people may have different experiences and thus different opinions. For example, one person bought a camera of a particular brand and had a very good experience with it. She naturally has a positive opinion or sentiment about the camera. However, another person who also bought a camera of the same brand had some issues with it because he might just be unlucky and got a defective unit. He thus has a negative opinion. Second, different people may see the same thing in different ways because everything has two sides. For example, when the price of a stock is falling, one person may feel very sad because he bought the stock when the price was high, but another person may be very happy because it is an opportunity to short sell the stock to make good profits. Furthermore, different people may have different interests and/or different ideologies.

2.2 OBJECTIVE OF THE PROJECT

Sentiment analysis or Opinion mining, as it is sometimes called, is one of many areas of computational studies that deal with opinion oriented natural language processing. Such opinion-oriented studies include among others, genre distinctions, emotion and mood recognition, ranking, relevance computations, perspectives in text, text source identification and opinion-oriented summarization. Sentiment analysis has turned out as an exciting new trend in social media with a gamut of practical applications that range from applications in business (marketing intelligence; product and service bench marking and improvement), applications as subcomponent technology. Sentiment is an attitude, thought, or judgment prompted by feeling. Sentiment analysis is also known as opinion mining. Sentiment analysis or opinion mining is one of the major tasks of NLP (Natural Language Processing). It has gained much attention in recent years. Internet is a resourceful place with respect to sentiment information. From a user's point of view, people are able to post their own content through various social media, such as forums, micro-blogs, or online social networking sites. From are searcher's perspective, many social media sites release their application programming interfaces (APIs), prompting data collection and analysis by researchers and developers. Sentiment Analysis, an area of Natural Language Processing (NLP), is used to classify the reviews using the sentiment of the words into positive or negative. Using the sentiment expressed in the words, opinions on any entity can be categorized into positive or negative. For example, the sentence, 'I am not excited by this product though it is quite cheap' expresses a negative sentiment about the product. The degree of the sentiment used is also taken into consideration. For example, 'I love this product' indicates amore positive sentiment than the sentence 'I like this product'. Apart from regular adjectives like 'good', 'bad' and 'very good', conjunctions like 'but', 'although", 'while' also have a say in the overall polarity of the sentence. There is a vast amount of information available on the Web which can assist individuals and organization in decision making processes but at the same time present many challenges as organizations and individuals attempt to analyse and comprehend the collective opinion of others. Unfortunately finding opinion sources, monitoring them and then analysing them are herculean tasks. It is not possible to manually find opinion sources online, extract sentiments from them and then to express them in a standard format. In recent years, the explosion of social networking sites, blogs and review sites provide a lot of information. Millions of people express uninhibited opinions about various product features and their nuances. This forms active feedback which is of importance not only to the companies developing the products, but also to their rivals and several other potential customers

2.3 BLOCK DIAGRAM OF PROJECT

Visualizing a complex system can be a good way to develop a more in-depth understanding of how it operates. Graphing or mapping out the system as a diagram may help you learn the different aspects of a system. One commonly used diagram is the block diagram, which provides a graphical overview of how the process flows together. In this article, we discuss what block diagrams are, how they work, their uses and how to design one.

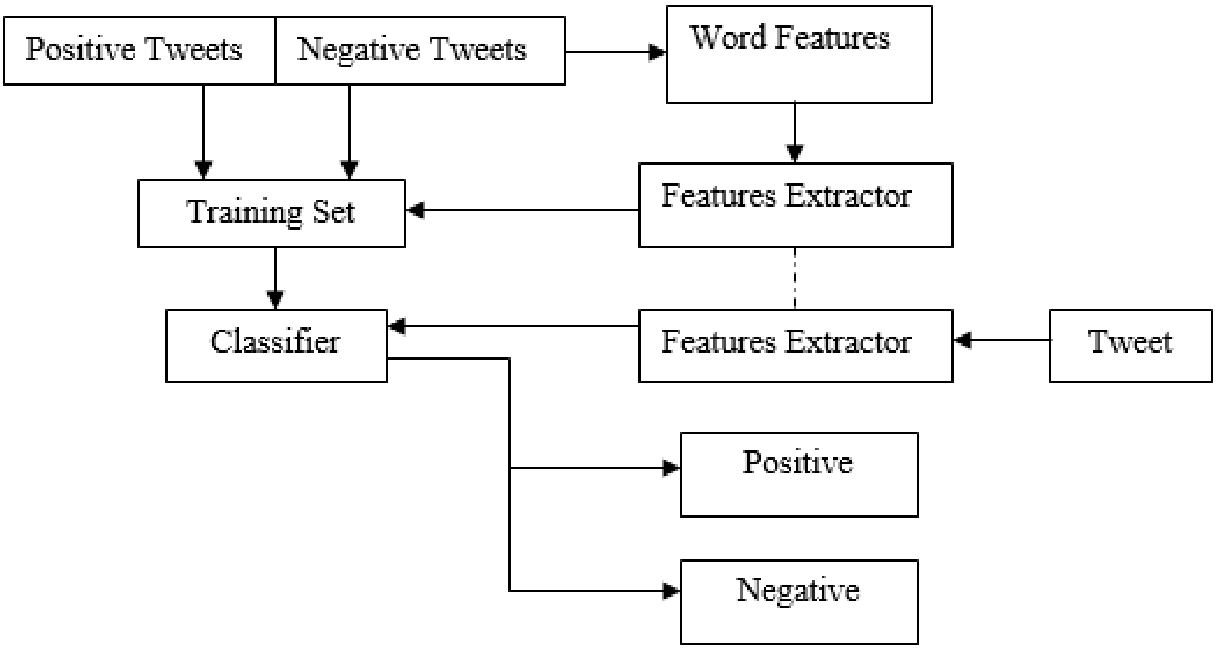

Fig. 2.3 Block diagram of sentiment analysis

2.4 MODULE REQUIREMENTS

The module requirements specification is the low-level design specification for the system. At this level, we will be concerned for the first time with implementing the system in a programming language metaphor. Each programming language creates a different metaphor. There is, for example, a C++ machine. This particular language metaphor is created through a very large library of runtime support services. There is a distinctly different metaphor created by the Java runtime environment. We will not concern ourselves with this low-level implementation detail until we actually begin to map our high-level design metaphor onto a particular language runtime environment. The module requirements specification is simply a low-level partition of the system functionality. The three central issues that emerge from the module description will be the structure of the system, the precise specification of data structures, and the data flow from one module to another. There must be, however, a one-to-one mapping between module specifications and code modules. Each module specification will result in exactly one code module. There will be no code modules that do not have design module specifications. A module is a distinct assembly of components that can be easily added, removed or replaced in a system. This is often used to make the software easier to use for a specific use case or to define where boundaries exist in a program.

Here the Modules are,

- Gathering data
- Pre-processing
- Classify and Visualize
- Building Model

2.5 HARDWARE SPECIFICATIONS

Processor	11th Gen Intel(R) Core (TM) i5-1135G7 @ 2.40GHz 2.42 GHz
Installed RAM	8.00 GB (7.70 GB usable)
System type	64-bit op erating system, x64-based processor

Fig. 2.5 Hardware specifications

2.6 SOFTWARE SPECIFICATIONS

- Google Chrome
- Microsoft Word
- Microsoft Excel

CHAPTER 3

LANGUAGE/TOOLS, DATA SET

CHAPTER 3

LANGUAGE/TOOLS, DATA SET

3.1 DATASET

The dataset for the sentiment analysis is taken from the website Kaggle. It contains tweets of Pfizer vaccine. It is in the format of CSV (comma separated value). It contains nearly 11000 datas of Pfizer vaccines.

3.2 LANGUAGE/TOOL DESCRIPTIONS

PYTHON

Here, I used python language for my project. Python is a high-level, general-purpose and a very popular programming language. Python programming language (latest Python 3) is being used in web development, Machine Learning applications, along with all cutting edge technology in Software Industry. Python Programming Language is very well suited for Beginners, also for experienced programmers with other programming languages like C++ and Java. This specially designed Python tutorial will help you learn Python Programming Language in most efficient way, with the topics from basics to advanced (like Web-scraping, Django, Deep-Learning, etc.) with examples. Rather than building all of its functionality into its core, Python was designed to be highly extensible via modules. This compact modularity has made it particularly popular as a means of adding programmable interfaces to existing applications. Van Rossum's vision of a small core language with a large standard library and easily extensible interpreter stemmed from his frustrations with ABC, which espoused the opposite approach. Python strives for a simpler, less-cluttered syntax and grammar while giving developers a choice in their coding methodology. In contrast to Perl's "there is more than one way to do it" motto, Python embraces a "there should be one—and preferably only one—obvious way to do it" philosophy. Alex Martelli, a Fellow at the Python Software Foundation and Python book author, wrote: "To describe something as 'clever' is *not* considered a compliment in the Python culture." Python's developers strive to avoid premature optimization and reject patches to non-critical parts of the CPython reference implementation that would offer marginal increases in speed at the cost of clarity. When speed is important, a Python programmer can move time-critical functions to extension modules written in languages such as C; or use PyPy, a just-in-time compiler. Cython is also available, which translates a Python script into C and makes direct C-level API calls into the Python interpreter. Python's developers aim for it to be fun to use. This is reflected in its name—a tribute to the British comedy group Monty Python —and in occasionally playful approaches to tutorials and reference materials, such as examples that refer to spam and eggs (a reference to a Monty Python sketch) instead of the standard foo, and bar. The programming language's name 'Python' came from the BBC Comedy series *Monty Python's Flying Circus*. Guido van Rossum thought he needed a name that was short, unique, and slightly mysterious, And so, he decided to name the programming language 'Python'.

3.2.1 PACKAGE/FUNCTIONS

PANDAS

Pandas is defined as an open-source library that provides high-performance data manipulation in Python. The name of Pandas is derived from the word **Panel Data**, which means **an Econometrics from Multidimensional data**. It is used for data analysis in Python and developed by **Wes McKinney** in **2008**.Data analysis requires lots of processing, such as **restructuring, cleaning** or **merging**, etc. There are different tools are available for fast data processing, such as **Numpy, Scipy, Cython**, and **Panda**. But we prefer Pandas because working with Pandas is fast, simple and more expressive than other tools.

SEABORN

Seaborn is an amazing visualization library for statistical graphics plotting in Python. It provides beautiful default styles and color palettes to make statistical plots more attractive. It is built on the top of matplotlib library and also closely integrated to the data structures from pandas. Seaborn aims to make visualization the central part of exploring and understanding data. It provides dataset-oriented APIs, so that we can switch between different visual representations for same variables for better understanding of dataset.

MATPLOTLIB

Matplotlib is an amazing visualization library in Python for 2D plots of arrays. Matplotlib is a multi-platform data visualization library built on NumPy arrays and designed to work with the broader SciPy stack. It was introduced by John Hunter in the year 2002. One of the greatest benefits of visualization is that it allows us visual access to huge amounts of data in easily digestible visuals. Matplotlib consists of several plots like line, bar, scatter, histogram etc.

NLTK

NLTK is a toolkit build for working with NLP in Python. It provides us various text processing libraries with a lot of test datasets. A variety of tasks can be performed using NLTK such as tokenizing, parse tree visualization, etc… In this article, we will go through how we can set up NLTK in our system and use them for performing various NLP tasks during the text processing step.

NUMPY

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays. It is the fundamental package for scientific computing with Python. Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data.

SCIKIT LEARN

Scikit-learn (Sklearn) is the most useful and robust library for machine learning in Python. It provides a selection of efficient tools for machine learning and statistical modelling including classification, regression, clustering and dimensionality reduction via a consistence interface in Python. This library, which is largely written in Python, is built upon NumPy, SciPy and Matplotlib.

REG EX

A RegEx, or Regular Expression, is a sequence of characters that forms a search pattern. RegEx can be used to check if a string contains the specified search pattern.

TEXTBLOB

TextBlob is an open-source Python library that is very easy to use for processing text data. It offers many built-in methods for common natural language processing tasks. Some of the tasks where I prefer to use it over other Python libraries are spelling correction, part of speech tagging, and text classification. But it can be used for various NLP tasks like:

1. Noun phrase extraction
2. Part of speech tagging
3. Sentiment Analysis
4. Text Classification
5. Tokenization
6. Word and phrase frequencies
7. Parsing
8. n-grams
9. Word inflexion

WORDCLOUD

Word Cloud is a data visualization technique used for representing text data in which the size of each word indicates its frequency or importance. Significant textual data points can be highlighted using a word cloud. Word clouds are widely used for analysing data from social network websites.

3.3 DATA SOURCE

To train a sentiment analysis model, we need machine learning techniques to help the model learn data patterns from specialized sentiment analysis datasets. Powered by artificial intelligence, when the sentiment analysis model is trained on these datasets, it knows how to behave when presented with new data in a similar vein; improving the accuracy of data analysis stage of sentiment analysis process. If you are a company in the hospitality industry, you will need a model that has been trained on datasets that are collected and tagged from the hospitality industry. And so is the case with all industry verticals. Such datasets need to be very wide in their scope of sentiment analysis applications and business cases. An efficiently trained sentiment model that can accurately analyse sentiment from text as well as videos, through video content analysis, is an invaluable asset for business intelligence. It can help you gain customer insights from not only reviews and surveys but also social platforms like YouTube, TikTok, Facebook, etc. Here the data is fetched from Kaggle which contains the tweets of Pfizer vaccines. It contains nearly 10000 data's. And it is in the format of csv (Comma Separated Value) file.

3.4 DATA STORAGE

The data is stored in the Windows10 MS Excel. It contains 16 columns of which text column is taken to proceed the project.

3.5 DATA CLEANING

Sentiment analysis allows you to mine emotions from data at scale and with precision. This is not to say that there are no sentiment analysis challenges. However, these can be easily navigated by ensuring that you have the right sentiment analysis platform and that data cleaning in sentiment analysis has been performed to the optimal level. Once these two aspects are taken care of, you can gather whatever insights you are looking for - whether from social media listening or mining Google and Amazon reviews for product research. However, to get to that point where you can conduct sentiment analysis, you need to make sure that your data is pristine. In data analytics terms, this means that there is no duplicate, incorrectly formatted, incomplete, corrupted, or simply, wrong data, in your dataset. While it may seem like an easy task when you are manually editing a couple of hundred comments, for example, TikTok insight but when you must assess numerous videos, say when doing an Instagram analysis, such scenarios mean videos with an aggregated sum of comments running into thousands. In such a case, you need an automated sentiment analysis tool. However, for that tool to give you accurate and high-precision results, you need to make sure that you have a high-quality dataset prepped for analysis. Incorrect sentiment analysis data preparation affects the algorithm and leads to incorrect analysis, even though at first glance it may all look in order. On the business front, this could mean a loss not only in resources and man-hours but also the risk of an expensive campaign gone wrong because it was built on incorrect insights. There are several measures that you can take to conduct data cleaning in sentiment analysis depending on the characteristic of the dataset you have. You can, however, establish a methodical approach such as below for sentiment analysis data preparation, and then use the same steps for all your future projects.

Step 1: Delete duplicate data

You should scan your dataset and delete anything that you think is irrelevant or duplicated. Removing duplicated data is very important especially if you are collating data from multiple sources either by using a web scraper, say for news monitoring, or if you are using your CRM tool and therefore gathering inter-department customer data. This could also be the case where you are analysing multiple videos based on a hashtag or keyword for Instagram social listening.

Step 2: Remove irrelevant items

Now that you have removed all duplicate data, you need to remove all items that are not relevant to your business objective. For this, you need to be properly aligned with the marketing goals for which you are conducting data cleaning in sentiment analysis. For example, if you are emotion mining for a voice of the customer analysis for a particular product line or age-group-based purchase behaviour, you need to remove any data that is not relevant to that particular objective. This means that even as you clean all your data manually and prepare the .csv file, ideally, you will have to conduct a separate analysis for each aspect of your business. This may seem like a long process but Repustate's sentiment analysis platform, Repustate IQ, solves this challenge by seamlessly categorizing all the data with the help of semantic clustering and giving aspect-based sentiment scores automatically for each data group or aspect in the data.

Step 3: Check for outlier data

Outlier data is one that is at the extreme ends of the median, meaning it may be either too low or too high. If it looks like an improper data entry, you have to be the judge of whether you want to remove it or leave it. Removing outliers may generally help in ensuring better insight results as the outlier may be just irrelevant data, but it is not necessary to do so. It is better to observe and then take the action of either deleting them or letting them stay on in the dataset. Get further details about the sentiment analysis datasets.

Step 4: Correct typos and structural mistakes

This may seem like a simple task while data cleaning in sentiment analysis but is a very important one. Once all the data has been sieved and you've got the first cut of the actual data you will be analysing, now is the time to correct any structural mistakes such as typos and inconsistencies in capitalizations, nomenclature, abbreviations, etc.

Step 5: Check for missing data

Missing data is never a good scenario but is not uncommon. You can solve this issue by either just removing all missing values from your analysis, or you can add a missing value based on assumption. Ofcourse, neither case is ideal but in the absence of data, they can be considered as options.

Step 6: Validate your data

Once you have undertaken all of the above-mentioned sentiment analysis data preparation steps, it is time to take stock of the end result that you have in front of you. Verify and validate your data to see if it makes sense in context of the overall objective of your marketing requirement. You can check to notice if there are any data deviations from the prescribed rules for its field as well. You must also verify if the analysis is giving you any trends that are not in sync with what you determined, which could mean that the quality of the data needs to be better. For example, this could be the case with surveys, so you may need to revisit your questionnaire.

CHAPTER 4

DATA GATHERING

CHAPTER 4

DATA GATHERING

4.1 OVERVIEW

The process of gathering and analysing accurate data from various sources to find answers to research problems, trends and probabilities, etc., to evaluate possible outcomes is Known as Data Collection. Keep scrolling to know more. Knowledge is power, information is knowledge, and data is information in digitized form, at least as defined in IT. Hence, data is power. But before you can leverage that data into a successful strategy for your organization or business, you need to gather it. That's your first step. So, to help you get the process started, we shine a spotlight on data collection. Our society is highly dependent on data, which underscores the importance of collecting it. Accurate data collection is necessary to make informed business decisions, ensure quality assurance, and keep research integrity. During data collection, the researchers must identify the data types, the sources of data, and what methods are being used. We will soon see that there are many different data collection methods. The data of Pfizer vaccines is collected from the source of Kaggle. It contains nearly 11000 real tweets datasets.

4.2 READING DATAFRAME

Pandas Data Frame is two-dimensional size-mutable, potentially heterogeneous tabular data structure with labelled axes (rows and columns). A Data frame is a two-dimensional data structure, i.e., data is aligned in a tabular fashion in rows and columns. Pandas Data Frame consists of three principal components, the **data**, **rows**, and **columns**. The dataset is readied through pandas.

4.3 DATA CLEANING

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset. When combining multiple data sources, there are many opportunities for data to be duplicated or mislabelled. If data is incorrect, outcomes and algorithms are unreliable, even though they may look correct. There is no one absolute way to prescribe the exact steps in the data cleaning process because the processes will vary from dataset to dataset. But it is crucial to establish a template for your data cleaning process so you know you are doing it the right way every time.

4.3.1 REMOVING UNWANTED DATA

Data's such as tweets with URL, lowercase conversions, hashtags, punctuations, stop words are removed by creating a function which is said to be pre-processed.

4.3.2 DROP DUPLICATE METHOD

Pandas drop_duplicates() method helps in removing duplicates from the Pandas Data frame In python.

Syntax: DataFrame.drop_duplicates(subset=None, keep='first', in place=False)

4.3.3 STEMMING

Stemming is the process of producing morphological variants of a root/base word. Stemming programs are commonly referred to as stemming algorithms or stemmers. A stemming algorithm reduces the words "chocolates", "chocolatey", "Choco" to the root word, "chocolate" and "retrieval", "retrieved", "retrieves" reduce to the stem "retrieve". Stemming is an important part of the pipelining process in Natural language processing. The input to the stemmer is tokenized words. How do we get these tokenized words? Well, tokenization involves breaking down the document into different words.

4.4 DATA ANALYSIS

Data analysis is the process of collecting, modelling and analysing data to extract insights that support decision-making. There are several methods and techniques to perform analysis depending on the industry and the aim of the investigation.

CHAPTER 5
POLARITY AND SENTIMENT

CHAPTER 5

POLARITY AND SENTIMENT

5.1 OVERVIEW

Polarity and sentiment of the tweets are checked and are added to the data frame. Then the positive, negative, neutral sentiments are categorized and visualized through pie chart and bar chart. It is the expression that determines the sentimental aspect of an opinion.

5.2 POLARITY CHECKING

In textual data, the result of sentiment analysis can be determined for each entity in the sentence, document or sentence. The sentiment polarity can be determined as positive, negative and neutral. A basic task in sentiment analysis classifying whether the expressed opinion in a document, a sentence or an entity feature/aspect is positive, negative, or neutral. Sentiment polarity for an element defines the orientation of the expressed sentiment, i.e., it determines if the text expresses the positive, negative or neutral sentiment of the user about the entity in consideration.

5.3 VISUALIZATION OF SENTIMENTS

The design goal of a large category of sentiment visualization techniques is to visually summarize the change of sentiments over time regarding to in a given streaming dataset such as news corpus, review comments, and Twitter streams. This goal can be approached, as by showing the sentiment dynamics in a time-series diagram where the time-series curve illustrates the change of sentiment scores computed across the entire dataset at different time points. However, this simple visualization is too abstract to display information details such as the causes behind the sentiment shifts. Therefore, many other highly advanced techniques have been introduced to illustrate and interpret the sentiment dynamics from different prospective. The sentimental data is visualized using bar chart and pie chart. And the positive, negative, neutral sentiment words are gathered through the word cloud and visualized.

CHAPTER 6

BUILDING MODEL

CHAPTER 6

BUILDING MODEL

6.1 OVERVIEW

The model is built using Logistic Regression and Support Vector Classifier. And it is compared in which the accuracy is most. Further to improve these algorithms accuracy hyperparameter tuning is done.

6.2 LOGISTIC REGRESSION

Logistic Regression is a classification that serves to solve the binary classification problem. The result is usually defined as 0 or 1 in the models with a double situation. Estimation is made by applying binary classification with Logistic Regression on the data allocated to training and test data in a data set below. First of all, Standardization for pre-processing will be applied, then training data will be trained with fit( ) and then it will be used to estimate test data with the predict( ) method.

6.3 SUPPORT VECTOR CLASSIFIER

SVM has defined input and output format. Input is a vector space and output are 0 or 1 (positive/negative). Text document in original form is not suitable for learning. They are transformed into format which matches into input of machine learning algorithm input. For this pre-processing on text documents is carried out. Then we carryout transformation. Each word will correspond to one dimension and identical words to same dimension. As mentioned, before we will see TF-IDF for this purpose. Now a machine learning algorithm is used for learning how to classify documents, i.e. creating a model for input-output mappings. SVM has been proved one of the powerful learning algorithms for text categorization. Text categorization systems may make mistakes. To compare different text classifiers for deciding which one is better, performance measures are used. Some of these measures the performance on one binary category, others aggregate per-category measures, to give an overall performance. TP, FP, TN, FN are the number of true/false positives/ negatives. The most important per-category measures for binary categories are,

- Precision: p=TP/(TP+FP)
- Recall: r= TP/(TP+FN)

The most important averages are: micro-average, which counts each document equally important, and macro-average, which counts each category equally important.

CHAPTER 7

RESULTS AND DISCUSSIONS

CHAPTER 7

RESULTS AND DISCUSSIONS

Sentiment analysis is a technique used to understand the emotional tone of the text. It can be used to identify positive, negative, and neutral sentiments in a piece of writing. This information can be useful for business owners who want to understand how their customers feel about their company. By understanding the sentiment of your customer's reviews and feedback, you can work to improve those areas that are causing dissatisfaction and increase loyalty among your customer bases. Marketers can use sentiment analysis to better understand customer feedback and adjust their strategies accordingly. Additionally, it can be used to determine whether a particular campaign or product resonates with customers in a positive or negative way. While it may seem like a complicated process, sentiment analysis is actually fairly straightforward

7.1INPUT AND OUTPUT SCREEN

7.1.1 INPUT

```
import pandas as pd
import numpy as np
import re
import seaborn as sns
import matplotlib.pyplot as plt
from matplotlib import style
style.use('ggplot')
from textblob import TextBlob
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
from nltk.corpus import stopwords
stop_words = set(stopwords.words('english'))
from wordcloud import WordCloud
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, ConfusionMatrixDisplay
```

```
df = pd.read_csv('vaccination_tweets.csv')
```

```
df.head()
```

Fig. 7.1.1 Importing required libraries

```
def data_processing(text):
    text = text.lower()
    text = re.sub(r"https\S+|www\S+https\S+", '',text, flags=re.MULTILINE)
    text = re.sub(r'\@w+|\#','',text)
    text = re.sub(r'[^\w\s]','',text)
    text_tokens = word_tokenize(text)
    filtered_text = [w for w in text_tokens if not w in stop_words]
    return " ".join(filtered_text)
```

```
text_df.text = text_df['text'].apply(data_processing)
```

```
text_df = text_df.drop_duplicates('text')
```

```
stemmer = PorterStemmer()
def stemming(data):
    text = [stemmer.stem(word) for word in data]
    return data
```

```
text_df['text'] = text_df['text'].apply(lambda x: stemming(x))
```

```
text_df.head()
```

Fig. 7.1.2 Data Processing is done

```
def sentiment(label):
    if label <0:
        return "Negative"
    elif label ==0:
        return "Neutral"
    elif label>0:
        return "Positive"
```

```
text_df['sentiment'] = text_df['polarity'].apply(sentiment)
```

```
text_df.head()
```

Fig 7.1.3 Sentiments are divided based on their categories

```
vect = CountVectorizer(ngram_range=(1,2)).fit(text_df['text'])
```

```
feature_names = vect.get_feature_names()
print("Number of features: {}\n".format(len(feature_names)))
print("First 20 features:\n {}".format(feature_names[:20]))
```

Fig 7.1.4 Features are extracted from the data

```
X = text_df['text']
Y = text_df['sentiment']
X = vect.transform(X)
```

```
x_train, x_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

```
print("Size of x_train:", (x_train.shape))
print("Size of y_train:", (y_train.shape))
print("Size of x_test:", (x_test.shape))
print("Size of y_test:", (y_test.shape))
```

Fig 7.1.5 Data's are splited into test and trained data

```
logreg = LogisticRegression()
logreg.fit(x_train, y_train)
logreg_pred = logreg.predict(x_test)
logreg_acc = accuracy_score(logreg_pred, y_test)
print("Test accuracy: {:.2f}%".format(logreg_acc*100))
```

```
Test accuracy: 84.64%
```

```
print(confusion_matrix(y_test, logreg_pred))
print("\n")
print(classification_report(y_test, logreg_pred))
```

Fig 7.1.6 Data's are trained by Logistic Regression

```
from sklearn.model_selection import GridSearchCV
```

```
param_grid={'C':[0.001, 0.01, 0.1, 1, 10]}
grid = GridSearchCV(LogisticRegression(), param_grid)
grid.fit(x_train, y_train)
```

```
GridSearchCV(estimator=LogisticRegression(),
             param_grid={'C': [0.001, 0.01, 0.1, 1, 10]})
```

```
print("Best parameters:", grid.best_params_)
```

```
Best parameters: {'C': 10}
```

```
y_pred = grid.predict(x_test)
```

```
logreg_acc = accuracy_score(y_pred, y_test)
print("Test accuracy: {:.2f}%".format(logreg_acc*100))
```

```
Test accuracy: 85.92%
```

Fig 7.1.7. Data’s are trained by GridSearchCV (one of hyparameter tuning method)

```
from sklearn.svm import LinearSVC
```

```
SVCmodel = LinearSVC()
SVCmodel.fit(x_train, y_train)
```

```
LinearSVC()
```

```
svc_pred = SVCmodel.predict(x_test)
svc_acc = accuracy_score(svc_pred, y_test)
print("test accuracy: {:.2f}%".format(svc_acc*100))
```

```
test accuracy: 87.34%
```

Fig 7.1.8 Data’s are trained by SVC algorithm

```
grid = {
    'C':[0.01, 0.1, 1, 10],
    'kernel':["linear","poly","rbf","sigmoid"],
    'degree':[1,3,5,7],
    'gamma':[0.01,1]
}
grid = GridSearchCV(SVCmodel, param_grid)
grid.fit(x_train, y_train)
```

```
GridSearchCV(estimator=LinearSVC(), param_grid={'C': [0.001, 0.01, 0.1, 1, 10]})
```

```
print("Best parameter:", grid.best_params_)
```

```
Best parameter: {'C': 10}
```

```
y_pred = grid.predict(x_test)
```

```
logreg_acc = accuracy_score(y_pred, y_test)
print("Test accuracy: {:.2f}%".format(logreg_acc*100))
```

```
Test accuracy: 87.58%
```

Fig 7.1.9 Data’s are trained using tuned SVC

7.2.1 OUTPUT

	id	user_name	user_location	user_description	user_created	user_followers	user_friends	user_favourites	user_verified	date	text	hashtags	source	retweets
0	1340539111971516416	Rachel Roh	La Crescenta-Montrose, CA	Aggregator of Asian American news; scanning di...	2009-04-08 17:52:46	405	1692	3247	False	2020-12-20 06:06:44	Same folks said daikon paste could treat a cyt...	['PfizerBioNTech']	Twitter for Android	0
1	1338158543359250433	Albert Fong	San Francisco, CA	Marketing dude, tech geek, heavy metal & '80s ...	2009-09-21 15:27:30	834	666	178	False	2020-12-13 16:27:13	While the world has been on the wrong side of ...	NaN	Twitter Web App	1
2	1337858199140118533	eliLTEU👌	Your Bed	heil, hydra 🖐☺	2020-06-25 23:30:28	10	88	155	False	2020-12-12 20:33:45	#coronavirus #SputnikV #AstraZeneca #PfizerBio...	['coronavirus', 'SputnikV', 'AstraZeneca', 'Pf...	Twitter for Android	0
3	1337855739918835717	Charles Adler	Vancouver, BC - Canada	Hosting "CharlesAdlerTonight" Global News Radi...	2008-09-10 11:28:53	49165	3933	21853	True	2020-12-12 20:23:59	Facts are immutable, Senator, even when you're...	NaN	Twitter Web App	446
4	1337854064604966912	Citizen News Channel	NaN	Citizen News Channel bringing you an alternati...	2020-04-23 17:58:42	152	580	1473	False	2020-12-12 20:17:19	Explain to me again why we need a vaccine @Bor...	['whereareallthesickpeople', 'PfizerBioNTech']	Twitter for iPhone	0

Fig. 7.2.1 Data’s are readied through pandas data frame

```
Index(['id', 'user_name', 'user_location', 'user_description', 'user_created',
       'user_followers', 'user_friends', 'user_favourites', 'user_verified',
       'date', 'text', 'hashtags', 'source', 'retweets', 'favorites',
       'is_retweet'],
      dtype='object')
```

Fig 7.2.2 Columns in the dataset are labelled

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11020 entries, 0 to 11019
Data columns (total 1 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   text    11020 non-null  object
dtypes: object(1)
memory usage: 86.2+ KB
```

Fig 7.2.3 Non null texts contains 11020 objects

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 10543 entries, 0 to 11019
Data columns (total 1 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   text    10543 non-null  object
dtypes: object(1)
memory usage: 164.7+ KB
```

Fig 7.2.4 After pre-processing the text data’s reduced to the rate of 10543 objects

	text	polarity	sentiment
0	folks said daikon paste could treat cytokine s...	0.000	Neutral
1	world wrong side history year hopefully bigges...	-0.500	Negative
2	coronavirus sputnikv astrazeneca pfizerbiontec...	0.000	Neutral
3	facts immutable senator even youre ethically s...	0.100	Positive
4	explain need vaccine borisjohnson matthancock ...	0.000	Neutral
5	anyone useful adviceguidance whether covid vac...	0.400	Positive
6	bit sad claim fame success vaccination patriot...	-0.100	Negative
7	many bright days 2020 best 1 bidenharris winni...	0.675	Positive
8	covid vaccine getting covidvaccine covid19 pfi...	0.000	Neutral
9	covidvaccine states start getting covid19vacci...	0.000	Neutral

Fig 7.2.5 All sentiments and its polarity are added to the data frame

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f9d6018ec50>
```

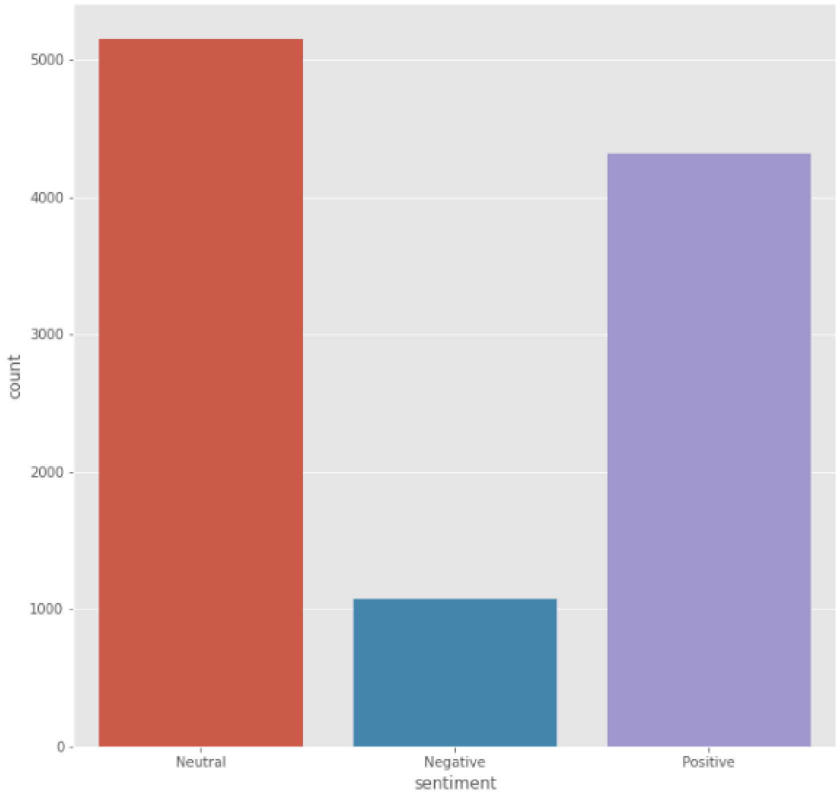

Fig 7.2.6 Bar chart represents different types of sentiments

```
Text(0.5, 1.0, 'Distribution of sentiments')
```

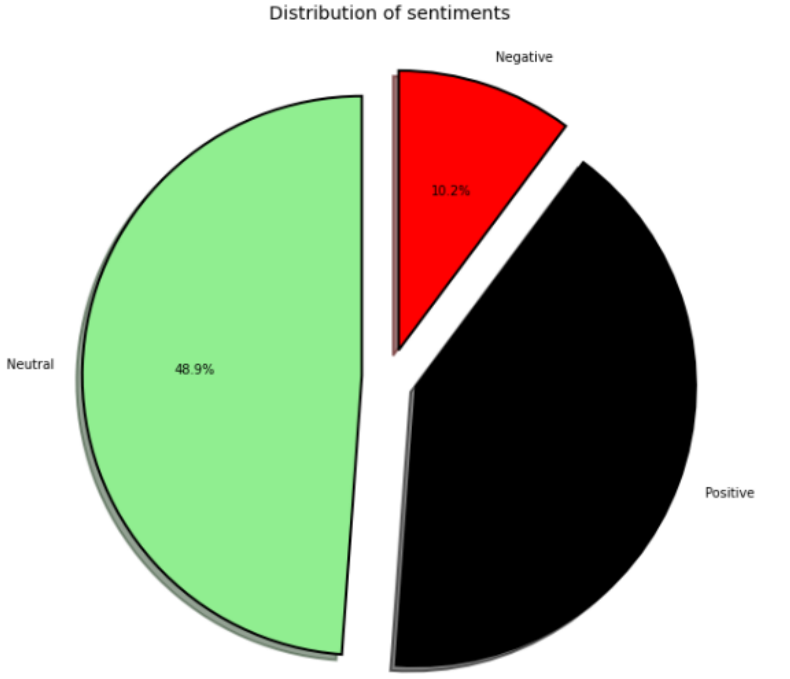

Fig 7.2.7 Pie chart represents the sentiments

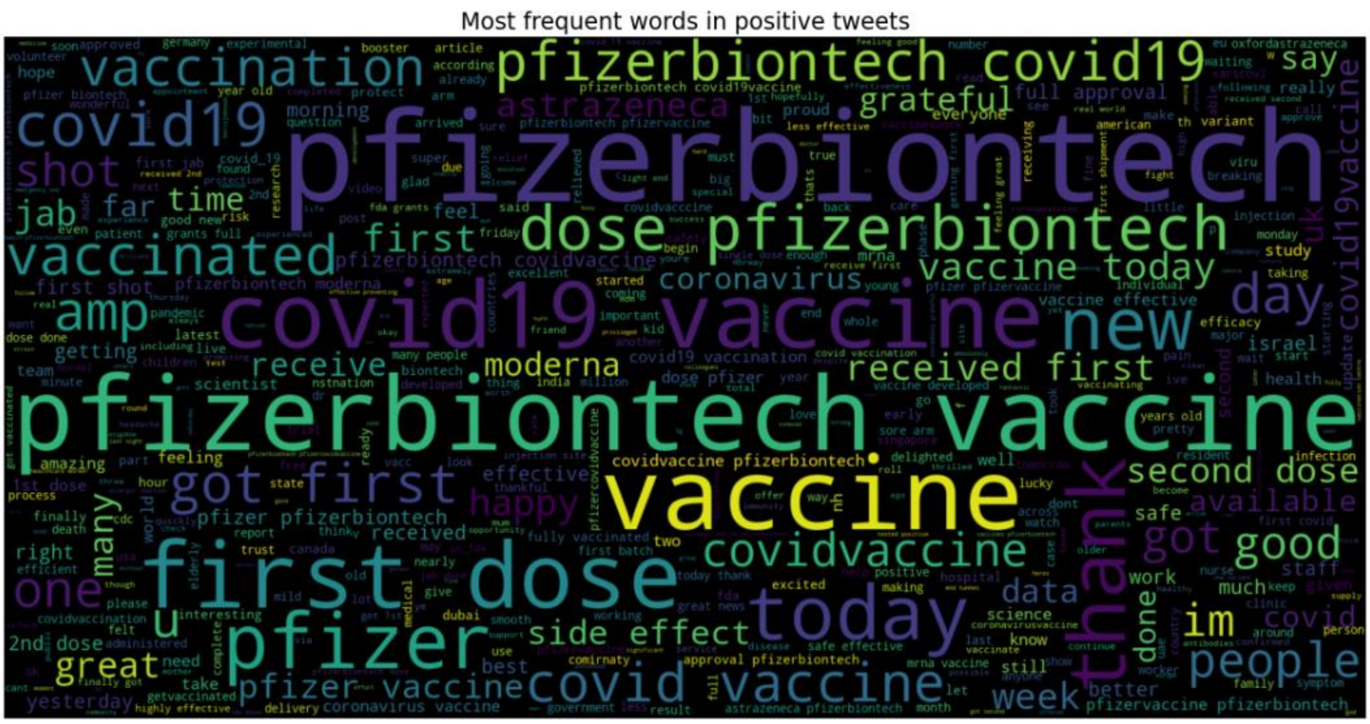

Fig 7.2.8 Word cloud created based on positive tweets

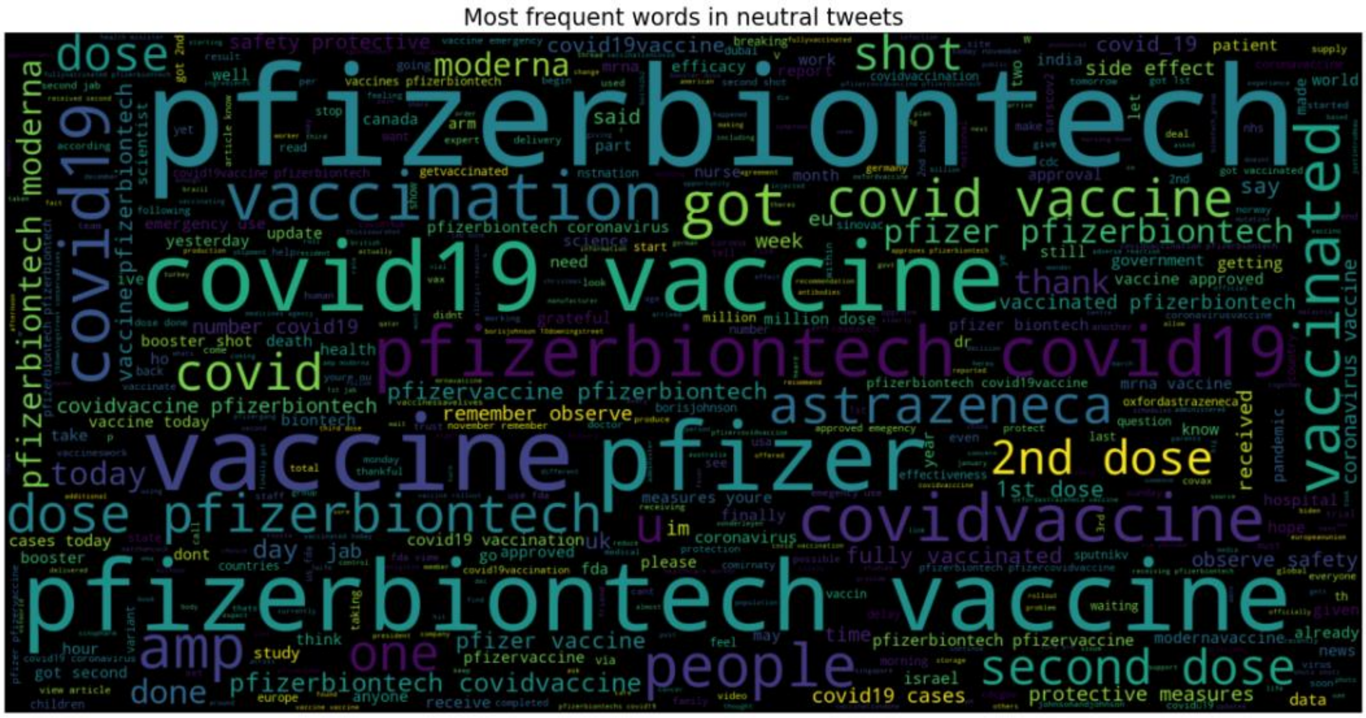

Fig 7.2.9 Word cloud created based on neutral tweets

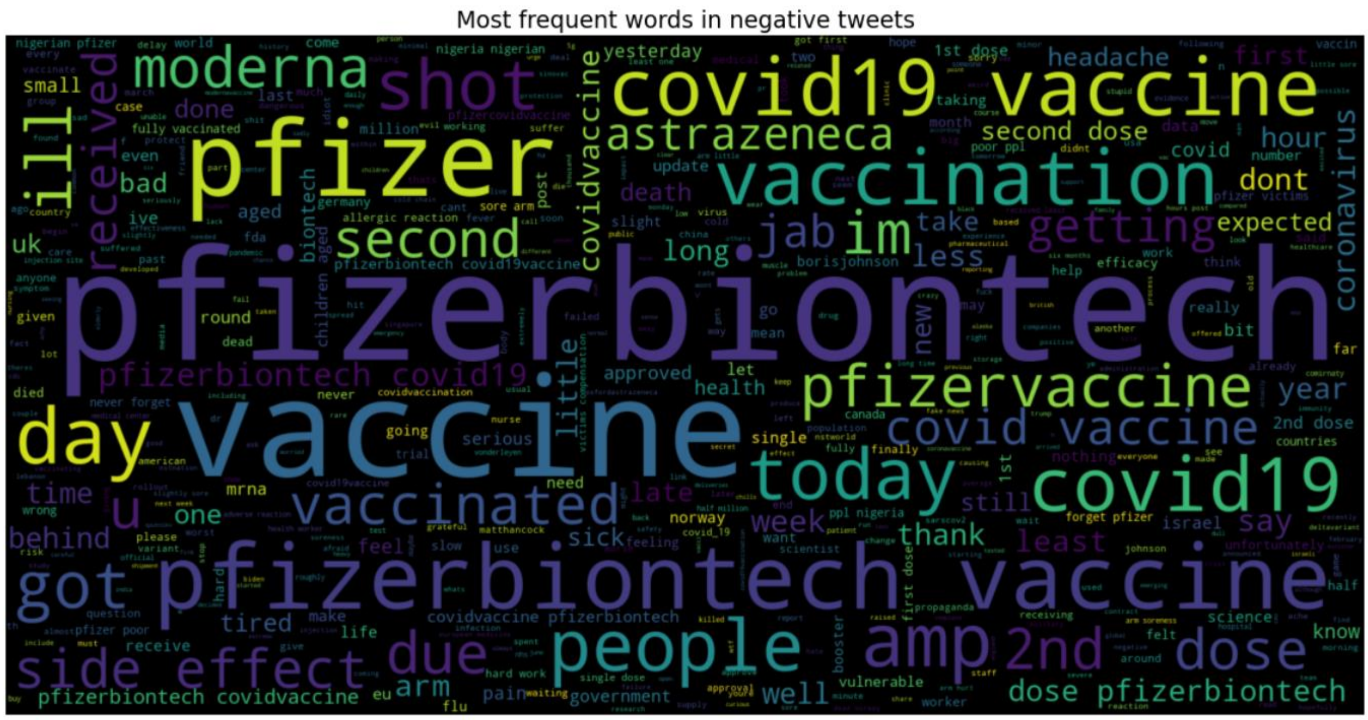

Fig. 7.2.10 Word cloud created based on negative tweets

```
logreg_acc = accuracy_score(y_pred, y_test)
print("Test accuracy: {:.2f}%".format(logreg_acc*100))
```

```
Test accuracy: 87.58%
```

```
print(confusion_matrix(y_test, y_pred))
print("\n")
print(classification_report(y_test, y_pred))
```

```
[[ 105   87   34]
 [   7 1005    9]
 [  14  111  737]]


              precision    recall  f1-score   support

    Negative       0.83      0.46      0.60       226
     Neutral       0.84      0.98      0.90      1021
    Positive       0.94      0.85      0.90       862

    accuracy                           0.88      2109
   macro avg       0.87      0.77      0.80      2109
weighted avg       0.88      0.88      0.87      2109
```

Fig. 7.2.11 The final highest accuracy using tuned SVM algorithm

7.2 PERFORMANCE EVALUATION

The function Logistic Regression will use the predict function and provide the accuracy of the model on the given testing data set. By using Logistic Regression, the model acquired 84.64 % and by doing hypermeter tuning the accuracy is build to 85.92%. And by using the function Support Vector Classifier the model acquired 87.34 %, and to improve the accuracy hypermeter tuning is done and it gives the accuracy as 87.58 %. It is found that the hyperparameter tuning in Support Vector Classifier model acquires the highest accuracy. Hence it is found to be the best model that fits for Sentimental Analysis.

7.3 DATA VISUALIZATION

Data visualization is a graphical representation of quantitative information and data by using visual elements like graphs, charts, and maps. Data visualization convert large and small data sets into visuals, which is easy to understand and process for humans. Data visualization tools provide accessible ways to understand outliers, patterns, and trends in the data. In the world of Big Data, the data visualization tools and technologies are required to analyse vast amounts of information. Data visualizations are common in your everyday life, but they always appear in the form of graphs and charts. The combination of multiple visualizations and bits of information are still referred to as Infographics. Data visualizations are used to discover unknown facts and trends. You can see visualizations in the form of line charts to display change over time. Bar and column charts are useful for observing relationships and making comparisons. A pie chart is a great way to show parts-of-a-whole. And maps are the best way to share geographical data visually. Today's data visualization tools go beyond the charts and graphs used in the Microsoft Excel spreadsheet, which displays the data in more sophisticated ways such as dials and gauges, geographic maps, heat maps, pie chart, and fever chart.

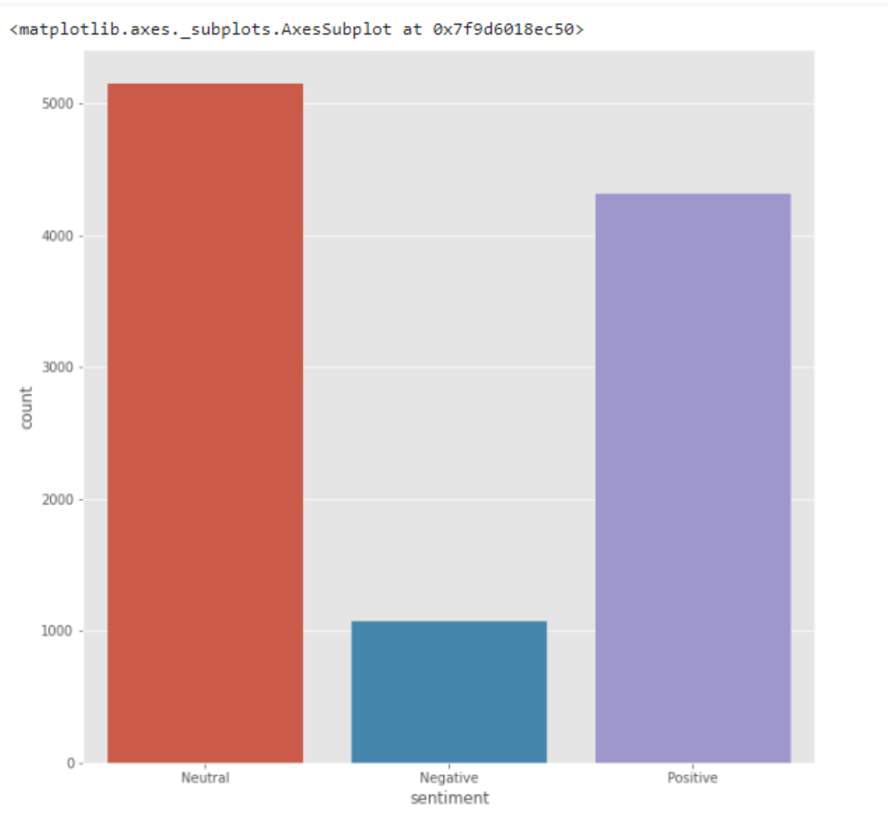

Fig. 7.3.1 Categorization of Sentiments

Text(0.5, 1.0, 'Distribution of sentiments')

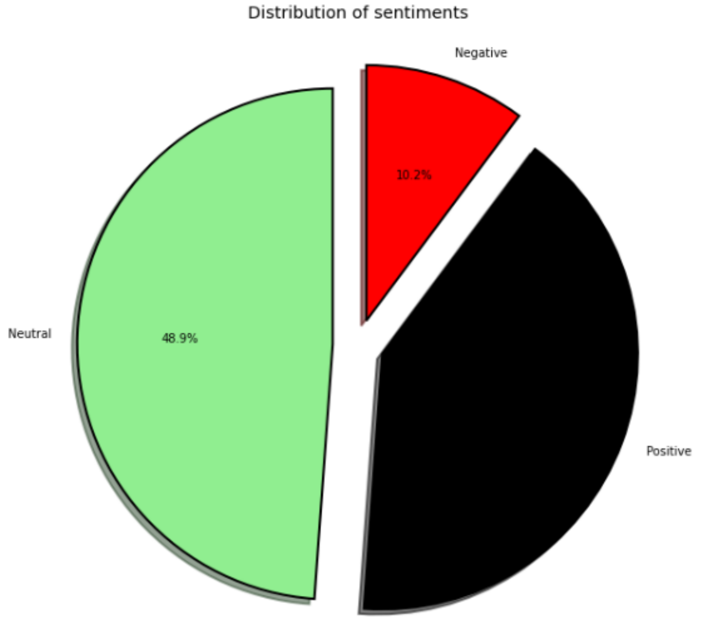

Fig. 7.3.2 Pie Chart represents Sentiment

<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7fd7c740ef10>

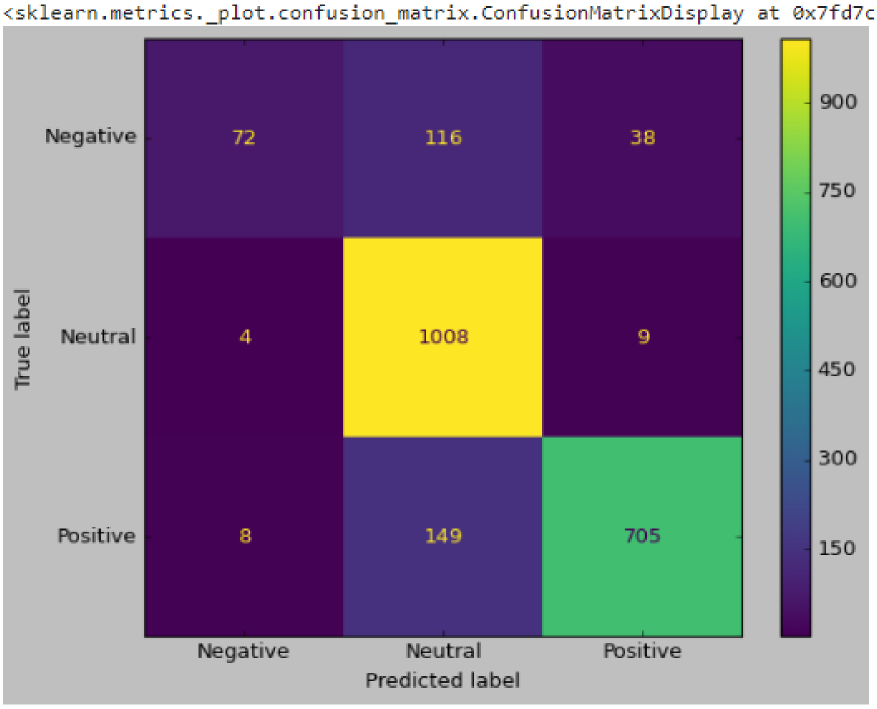

Fig. 7.3.3 Heat map shows the correlation between sentiments

7.4 COMPARISON OF ALGORITHM

SVM tries to find the “best” margin (distance between the line and the support vectors) that separates the classes and this reduces the risk of error on the data, while logistic regression does not, instead it can have different decision boundaries with different weights that are near the optimal point. SVM works well with unstructured and semi-structured data like text and images while logistic regression works with already identified independent variables. SVM is based on geometrical properties of the data while logistic regression is based on statistical approaches. The risk of overfitting is less in SVM, while Logistic regression is vulnerable to overfitting.

MODEL	**ACCURACY**
Logistic Regression	84.64%
Tuned Logistic Regression	85.92%
SVC	87.34%
Tuned SVC	87.58%

Fig. 7.4 Model accuracy

CHAPTER 8

CONCLUSION AND FUTURE WORK

CHAPTER 8
CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

In this study, it is analysed tweets related to Pfizer vaccines. It observed various patterns for the number of tweets, the vaccine ratings and sentiments of their ratings. Due to the fact that tweets often include the title of the vaccines, I created two distinct datasets: one with the title of the article in the tweets and one without. I built machine learning models to predict the sentiment of the tweet for a given research article. Ddeveloped classification models for two class and three class labels (positive, neutral, or negative). The tree-based classifiers generally outperformed other classifiers. The era of getting valuable insights from surveys and social media has peaked due to the advancement of technology. Therefore, it is time for your business to be in touch with the pulse of what your customers are feeling. Companies are using intelligent classifiers like contextual semantic search and sentiment analysis to leverage the power of data and get the deepest insights. Formulate business strategies, exceed customer expectations, generate leads, build marketing campaigns, and open new avenues for growth through natural language processing solutions.

8.2 FUTURE ENHANCEMENT

In the future, sentiment analysis will delve deeper, beyond the concept of the number of likes, comments, and shares in a post, to reach and comprehend the significance of social media conversations and what they reveal about consumers. As a result, sentiment analysis tools like Bytes View are becoming necessary for these businesses to survive in such a competitive market. Even today, most of the sentiment analysis in any project is carried out almost entirely by corporations and brands using data from social media, survey responses, and other sources of user-generated content. And, by examining and evaluating customer sentiments with such tools, these brands can gain an in-depth understanding of consumer behaviours and, as a result, better serve their audiences with the products, services, and experiences they provide.

CHAPTER 9

REFERENCE

CHAPTER 9

REFERENCES

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LIST OF FIGURES USED